

# Taha Bekar

Robotics Software Engineer — Simulation — Perception — Autonomous Systems  
mtahabekar@gmail.com — +1 (838) 262 0019 — LinkedIn — Github — Portfolio

## Education

|                                                    |          |                                             |                |
|----------------------------------------------------|----------|---------------------------------------------|----------------|
| <b>Rensselaer Polytechnic Institute</b>            | Troy, NY | <b>Middle East Technical University</b>     | Ankara, Turkey |
| M.S. Electrical, Computer, and Systems Engineering |          | B.S. Electrical and Electronics Engineering |                |
| 2024 – May 2026                                    |          | 2018 – 2024                                 |                |
| Focus: Autonomy, Robotics, and Control             |          | GPA: 3.56 / 4.00                            |                |

## Research Experience

### Deformable Object Manipulation Research – RPI

2025 – Present

Research Assistant / Master's Project

- Developing a centralized PPO-based reinforcement learning framework in NVIDIA Isaac Lab for coordinated dual-arm deformable-object transport, with continuous Cartesian control and privileged robot/object state observations.
- Building Scrub Nurse robot simulation on Isaac sim (Kinova JACO2 arm and Clearpath Dingo base) for medical tool picking using Isaac GROOT Vision-Language-Action (VLA) model.
- Developed a Python-based, ROS2-compatible deformable-object simulation framework in NVIDIA Isaac Sim using Newton Style3D for real-time cloth manipulation with anisotropic material, contact, friction, damping, and bending parameters.
- Built an obstacle-aware cloth manipulation pipeline in Python with kinematic corner pinning, simulated cloth state publishing, ROS2 command interfaces, obstacle-distance computation, and closed-loop follower-corner control.
- Integrated a Control Barrier Function safety controller in Python for cloth-obstacle avoidance in contact-rich manipulation scenarios.
- Developed a real-time ROS2 perception and state estimation pipeline in Python for planar deformable objects using RGB-D data from an end-effector-mounted Azure Kinect, ArUco marker tracking, and Kalman filtering; deployed on a Kinova JACO2 arm and Clearpath Dingo mobile base for real-to-sim calibration.

## Work Experience

### Kuartis

Ankara, Turkey

Computer Vision / Machine Learning Engineer

Feb 2024 – Jun 2024

- Developed and benchmarked real-time, production-style C++ implementations of LiDAR point cloud registration algorithms, including ICP, NDT, GICP, and FAST-GICP, for SLAM and mapping systems.
- Implemented real-time LiDAR-IMU sensor fusion pipelines in C++ by combining LiDAR odometry with IMU odometry and evaluating tightly coupled and loosely coupled fusion strategies for improved localization robustness.
- Evaluated C++ implementations of SLAM frameworks including LOAM, LIO-SAM, and FAST-LIO for localization accuracy, runtime performance, and computational efficiency.
- Improved LIO-SAM localization accuracy by 6% by integrating GICP-based scan matching in C++ into the GTSAM factor-graph optimization framework.

### GT-ARC Gemeinnützige GmbH, DAI Labor, TU Berlin

Berlin, Germany

Software Engineer Intern

Jul 2023 – Sep 2023

- Developed a production-oriented C++ LiDAR-based free parking space detection algorithm for autonomous vehicles using ROS, Autoware, and HD Maps.
- Implemented geometric intersection analysis in C++ between detected vehicles and mapped parking regions, achieving 86% free space detection accuracy.

### Analog Devices

Istanbul, Turkey

Software Engineer Intern

Jul 2022 – Aug 2022

- Developed production-level C firmware for a PWM controller on the MAX32666 microcontroller using hardware timer peripherals; the implementation was incorporated into the official microcontroller firmware.
- Built a GUI interface for validating PWM signals and implemented automated unit tests using Google Test for the production firmware.

## Projects

### Autonomous UGV Mapping System

- Converted a 2D LiDAR into a 3D scanning system via servo rotation to enable volumetric mapping.
- Implemented OctoMap mapping using ICP (Iterative Closest Point) registration and AMCL (Adaptive Monte Carlo Localization) algorithms in ROS. Achieved accuracy levels of 95% and 75% for mapping and localization, respectively.

### Lane Detection System

- Implemented classical vision-based lane detection pipelines in C++ and MATLAB using Canny edge detection and the Hough transform.

## Technical Skills

**Programming:** Python, C++17, C, MATLAB

**Robotics:** ROS, ROS2, NVIDIA Isaac Sim, NVIDIA Isaac Lab, PhysX, Newton, Gazebo, 3D Kinematics, SLAM, Reinforcement Learning, PPO, Isaac GROOT, VLA

**Perception:** RGB-D Perception, LiDAR Point Cloud Processing, Sensor Fusion, TF, OptiTrack

**Libraries:** PCL, OpenCV, PyTorch, NumPy, rclcpp, tf2, Eigen, GTSAM, Google Test

**Tools:** Linux, Git, Docker